MOMENTUM BUILDER

Strategy Blueprint & Research Guide

How the Score Works · The 5 Indicators · The 5 Strategies

NSE (India ■) & NYSE/NASDAQ (USA $) — Long-Term Momentum Investing

Version 2.0 · Built with Python + Flask + Yahoo Finance · May 2026

5 Indicators	**5 Strategies**	**2 Markets**	**81–100 Elite Score**

Purpose of this document: A complete reference explaining how every momentum score is calculated, which academic papers back each indicator, how the 5 portfolio strategies work, and when to buy or sell. Designed to be read once and referred to monthly.

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1. WHAT IS MOMENTUM INVESTING?

The Core Idea

Momentum investing is built on one simple observation, proven across markets worldwide: **stocks that have performed well in the recent past tend to continue performing well in the near future.** Conversely, stocks that have performed poorly tend to continue underperforming. This effect is not random — it is one of the most replicated findings in financial economics.

The academic term is the "momentum anomaly" — it is called an anomaly because it contradicts the Efficient Market Hypothesis (EMH), which states that past prices cannot predict future returns. Yet momentum has been proven to work in over 40 countries, across stocks, bonds, currencies, and commodities, going back as far as 1801.

Key Finding: Jegadeesh & Titman (1993) showed that buying the top-performing stocks over the past 12 months and selling the worst-performing generated approximately **1–2% monthly excess returns** over a 3–12 month holding period. This was confirmed out-of-sample by Fama & French (1996) — the same researchers who built the Efficient Market Hypothesis framework.

Why Does Momentum Work?

Behavioural finance gives us three main explanations:

Cause	Explanation
Under-reaction	Investors are slow to incorporate good news. The stock rises gradually over months as more investors notice
Herding	Once a trend is established, more investors pile in (trend-following), amplifying the move further.
Anchoring	Investors anchor to recent highs/lows. Near a 52-week high, they hesitate to buy — then when the stock brea

Why Monthly Data?

This tool uses **monthly closing prices** rather than daily data for all indicator calculations. Monthly data removes daily noise, reduces false signals, and is far more suitable for long-term investors who are not watching screens all day. Academic momentum research (Jegadeesh & Titman, Fama & French, Antonacci) is all conducted on monthly data. Daily signals require daily monitoring — monthly signals require only monthly rebalancing.

2. THE SCORING SYSTEM — 0 TO 100

Every stock scanned by Momentum Builder receives a single composite score between 0 and 100. This score is a **weighted average of 5 independent momentum indicators**, each scored on a 0–100 scale.

The Formula

Indicator	Weight	Max Contribution
1. 12-1 Month Price Momentum	30%	30 points
2. Monthly RSI (14-period)	20%	20 points
3. Monthly MACD (12-26-9)	20%	20 points
4. 52-Week High Proximity	15%	15 points
5. 6M vs 12M SMA Cross	15%	15 points
TOTAL	**100%**	**100 points**

Final Score = (Ind1 × 0.30) + (Ind2 × 0.20) + (Ind3 × 0.20) + (Ind4 × 0.15) + (Ind5 × 0.15)

Score Classification

Score Range	Class	Meaning	Action
81 – 100	**ELITE / Very Strong**	All or most indicators firing strongly	SCREENER PICKS THESE
65 – 80	**Strong**	Good momentum, most indicators positive	Consider building position
50 – 64	**Moderate**	Mixed signals, some indicators positive	Wait for confirmation
35 – 49	**Weak**	Momentum fading, mostly negative signals	Caution — reduce exposure
0 – 34	**Very Weak**	Negative momentum, avoid	Avoid or exit

3. INDICATOR 3 — 12-1 MONTH PRICE MOMENTUM (30%)

12-1 Month Price Momentum

Weight: 30%

Academic Backing: Jegadeesh & Titman (1993), Fama & French (1996)

Data Used: Monthly closing prices — 13 months of data required

Formula

Momentum Return = (Price 1 month ago ÷ Price 13 months ago – 1) × 100

How the Score Is Calculated (0–100)

• The return over the past 12 months is calculated, but the most recent month is skipped.

• Skipping the last month avoids the "1-month reversal anomaly" — stocks that rise sharply in one month often pull back the next.

• The raw return is then mapped to a 0–100 score: –30% maps to 0, +40% maps to 100.

• A 12-month return of +5% → Score ≈ 50. A return of +25% → Score ≈ 79.

• 6-month and 3-month returns are also calculated for additional context in the detail view.

Signal Interpretation

Score Range	Raw Return	Signal
90 – 100	> +20%	Very Strong Momentum
70 – 89	+10% to +20%	Strong Momentum
50 – 69	0% to +10%	Moderate Momentum
30 – 49	–10% to 0%	Weak / Fading Momentum
0 – 29	< –10%	Negative Momentum

Why 30% Weight?

Price momentum has the strongest and most replicated academic evidence of any factor in finance. Proven in 40+ countries over 200+ years. It directly measures what we are trying to capture — sustained price appreciation. All other indicators are derived from price; this one IS the price signal. Hence 30% — the highest weight.

4. INDICATOR 4 — MONTHLY RSI (14-PERIOD) (20%)

Monthly RSI (14-period)

Weight: 20%

Academic Backing: J. Welles Wilder (1978) — Validated for long-term by multiple studies

Data Used: Monthly closing prices — 15+ months of data required

Formula

RSI = 100 – [100 ÷ (1 + Average Gain ÷ Average Loss)] over 14 monthly periods

How the Score Is Calculated (0–100)

• RSI is calculated on monthly closing prices using Wilder's smoothing (EWM with com=13).

• RSI naturally produces a 0–100 value — so the RSI value IS the score directly.

• RSI > 50 = more up-months than down-months over 14 months = bullish territory.

• RSI > 60 = strong consistent uptrend. RSI > 70 = very strong but watch for overbought.

• Rising RSI (this month > last month) is an additional positive signal noted in the detail view.

Signal Interpretation

RSI Score	Zone	Interpretation
70 – 100	Overbought / Very Strong	Strong trend but watch for reversal
60 – 69	Strong Momentum Zone	Ideal momentum range — sustained trend
50 – 59	Bullish Territory	Positive but moderate momentum
40 – 49	Bearish Territory	More down-months than up recently
0 – 39	Weak / Oversold	Negative momentum — avoid

Why 20% Weight?

RSI is a reliable trend-confirmation tool but is derived entirely from the same price data as Indicator 1. It adds smoothing and normalisation but does not introduce truly independent information. Given this overlap, 20% is appropriate — it confirms the momentum direction but does not add a new data dimension.

5. INDICATOR 5 — MONTHLY MACD (12-26-9) (20%)

Monthly MACD (12-26-9)

Weight: 20%

Academic Backing: Gerald Appel (1979) — Multiple quant studies confirm monthly MACD alpha

Data Used: Monthly closing prices — 35+ months of data required (26 + 9)

Formula

MACD Line = EMA(12) – EMA(26) | Signal Line = EMA(9) of MACD | Histogram = MACD – Signal

How the Score Is Calculated (0–100)

- MACD is calculated on monthly closes using exponential moving averages.
- The Histogram (MACD minus Signal) is the key value — positive = bullish, negative = bearish.
- Scoring logic: MACD above signal line = base score 65. Below = base score 35.
- If histogram is rising (momentum accelerating) → bonus up to +20 points.
- If histogram is falling (momentum decelerating) → penalty up to –15 points.
- Maximum score of 100 = MACD above signal AND histogram rising strongly.

Signal Interpretation

Condition	Score Range	Signal
MACD above + Histogram rising	75 – 100	Bullish & Accelerating — strongest
MACD above + Histogram falling	50 – 74	Bullish but Decelerating — caution
MACD below + Histogram rising	35 – 49	Bearish but Recovering — watch
MACD below + Histogram falling	0 – 34	Bearish & Weakening — avoid

Why 20% Weight?

MACD measures momentum acceleration — not just direction but whether the trend is speeding up or slowing down. A stock with MACD decelerating is often about to stall, even if price momentum looks good. However, like RSI, MACD is price-derived, so 20% weight is appropriate. Together RSI + MACD at 40% provide strong trend-quality filtering.

6. INDICATOR 6 — 52-WEEK HIGH PROXIMITY (15%)

52-Week High Proximity

Weight: 15%

Academic Backing: George & Hwang (2004) — "The 52-Week High and Momentum Investing"

Data Used: Monthly closing prices — 13 months of data required

Formula

Ratio = (Current Price ÷ 52-Week High) × 100 | Score = (Ratio × 70%) + (Position in Range × 30%)

How the Score Is Calculated (0–100)

• The 52-week high is the highest monthly High over the past 12 months (excluding current month).

• Ratio = how close is today's price to that 52-week high? E.g. 95% = only 5% below the high.

• Position in Range = where is the price within the 52-week High-Low range (0%=at low, 100%=at high).

• Final score = 70% weight on ratio + 30% weight on position in range.

• A stock AT its 52-week high scores ~100. A stock AT its 52-week low scores ~0.

Signal Interpretation

Score	Ratio to 52W High	Signal
90 – 100	≥ 95%	Near 52-Week High — Very Strong breakout potential
75 – 89	85–94%	Strong — approaching all-time range high
55 – 74	70–84%	Moderate — mid-range
35 – 54	55–69%	Weak — below midpoint of range
0 – 34	< 55%	Very Weak — near 52-week low

Why 15% Weight?

George & Hwang (2004) showed this signal outperforms traditional price momentum in many markets. It captures investor anchoring psychology — resistance near recent highs means that when a stock finally breaks through, the move is sustained. However, it is a narrower, more specific signal than price momentum itself, so 15% weight is appropriate.

7. INDICATOR 7 — 6M VS 12M SMA CROSS (15%)

6M vs 12M SMA Cross

Weight: 15%

Academic Backing: Mebane Faber (2007), Gary Antonacci (2014)

Data Used: Monthly closing prices — 12+ months of data required

Formula

6M SMA = Average of last 6 monthly closes | 12M SMA = Average of last 12 monthly closes | Spread = (6M SMA – 12M SMA) ÷ 12M SMA × 100

How the Score Is Calculated (0–100)

• Both the 6-month and 12-month Simple Moving Averages are calculated on monthly closes.

• Spread % shows how far apart the two SMAs are — positive = 6M above 12M = uptrend.

• Score maps the spread from –20% to +20% → 0 to 100.

• Bonus +5 points if price is above both SMAs. Penalty –5 points if below both.

• A "Golden Cross" (6M just crossed above 12M) is flagged as a special signal.

• A "Death Cross" (6M just crossed below 12M) is flagged as a warning.

Signal Interpretation

Score	Spread	Signal
80 – 100	> +10%	Very Strong Uptrend — Golden Cross zone
63 – 79	+3 to +10%	Strong Uptrend
53 – 62	0 to +3%	Mild Uptrend
43 – 52	–3 to 0%	Mild Downtrend
28 – 42	–10 to –3%	Downtrend — Death Cross zone
0 – 27	< –10%	Strong Downtrend — avoid

Why 15% Weight?

Faber (2007) showed monthly SMA systems beat buy-and-hold on a risk-adjusted basis with significantly reduced drawdowns. However, this indicator LAGS price momentum — the cross happens after the trend has already begun. It also overlaps with what Price Momentum already captures. Hence 15% weight — valuable confirmation, but not a primary signal.

8. WHY DIFFERENT WEIGHTS? (30/20/20/15/15)

The weights reflect two things: **(1) how much independent academic evidence exists** for each indicator, and **(2) how much independent information** each indicator adds beyond what the others already capture.

The Problem With Equal Weights

If we used equal weights (20% each), RSI + MACD + MA Cross would collectively receive 60% weight. But all three are calculated from the same monthly price data as Indicator 1. This means we would effectively be **triple-counting the same signal** while giving the one truly direct momentum measure only 20% weight.

Key Principle: The weight of each indicator should reflect its *marginal information content* — how much new, independent signal it adds beyond what we already know from other indicators. More independent evidence = higher weight.

Indicator	Data Source	Independence	Evidence Strength	Weight
Price Momentum	Raw price return	Direct — it IS the signal	200+ years, 40+ countries	**30%**
RSI	Price-derived	Overlaps with #1	Strong, widely validated	**20%**
MACD	Price-derived	Overlaps with #1 & #2	Strong — adds acceleration view	**20%**
52-Week High	Price anchoring	Different psychological angle	Strong but narrower scope	**15%**
MA Cross	Price-derived	Lagging — confirms #1 late	Good but slow signal	**15%**

9. THE SCREENER — SCORE 81 TO 100

Why 81 as the Threshold?

A score of 81+ means a stock is in the **top quintile (top 20%) of all momentum stocks**. Academic momentum research consistently shows that the strongest alpha comes from the top quintile — the top 20% of momentum stocks outperform the bottom 20% by 10–15% annually.

A score of 80 means all 5 indicators are in the "above average" zone. A score of 81+ means at least some indicators are firing *strongly*, not just moderately bullish. This filters out "borderline" stocks.

Universe Scanned

Exchange	Universe	Cap Categories	Approx Stocks
NSE (India)	Nifty 100 + Midcap 150 + Smallcap 250	Large / Mid / Small Cap	~494
NYSE/NASDAQ	S&P 500 + S&P 400 + S&P 600	Large / Mid / Small Cap	~869

Why Micro Cap is excluded: Micro cap stocks have insufficient liquidity — thin trading volume means prices can be stale, manipulated, or unrepresentative. Momentum signals require reliable price discovery, which micro caps often lack. All strategies in this tool are designed for liquid, investable stocks only.

Auto-Refresh

The screener runs automatically every 30 days via APScheduler. This matches the monthly rebalancing frequency used in academic momentum research. Running more frequently increases transaction costs without meaningfully improving returns. Running less frequently misses important momentum shifts.

10. STRATEGY 1 — DUAL MOMENTUM

■ Dual Momentum

Rebalance: **Monthly** | Max Positions: **15 (Bull) / 1 ETF (Bear)** | Risk: **Low-Medium**

Academic Backing: Gary Antonacci (2014) — 17.4% CAGR, –22.7% max drawdown (1974–2013)

Strategy Description

Dual Momentum combines two types of momentum: **Absolute Momentum** (is the overall market trending up?) and **Relative Momentum** (which stocks are the strongest within that trending market?). This is the only strategy that has a built-in market crash protection mechanism.

The strategy checks the equity market ETF (NIFTYBEES for NSE, SPY for NYSE) 12-month return every month. If positive = bull market, deploy capital into top stocks. If negative = bear market, protect capital in cash/safe ETF.

How Positions Are Built

→ Check equity ETF (NIFTYBEES / SPY) 12-month return.

→ BULL MODE (12M return > 0%): Invest ■100/$100 in each of the Top 15 momentum stocks from the screener.

→ BEAR MODE (12M return ≤ 0%): Invest entire capital in LIQUIDBEES (NSE) or BIL T-Bill ETF (NYSE).

→ Rebalance every month — swap new top-15 in, remove dropped-out stocks.

Entry & Exit Rules

Entry: Monthly — after screener runs, check market ETF signal, then allocate. **Exit:** Remove stocks that fall out of the top-15 ranking OR if market turns bearish. **Bear→Bull transition:** When market ETF 12M return crosses back above 0%, exit cash ETF and re-enter top-15 stocks.

11. STRATEGY 2 — QUALITY MOMENTUM 50

■ Quality Momentum 50

Rebalance: **Quarterly** | Max Positions: **50 stocks** | Risk: **Medium**

Academic Backing: Gray & Vogel (2016) Alpha Architect QMOM — 14.4% CAGR, –42% max drawdown

Strategy Description

Quality Momentum 50 selects the top 50 momentum stocks but adds a quality filter: only stocks with smooth, consistent price trends (MA score ≥ 55) are eligible. This removes "lottery ticket" momentum stocks that spiked briefly but have choppy price action — a concept from Alpha Architect's QMOM methodology.

A sector cap of 3 prevents any single sector from dominating the portfolio. Equal-weighting (■100/$100 per position) ensures no individual stock dominates returns.

How Positions Are Built

→ Sort all elite screener stocks (score 81+) by final momentum score, highest first.

→ Apply quality filter: MA momentum score must be ≥ 55 (smooth, consistent uptrend).

→ Apply sector cap: maximum 3 stocks per sector.

→ Take the top 50 stocks that pass both filters.

→ If fewer than 50 pass, fill remaining slots with next-ranked stocks ignoring quality filter.

Entry & Exit Rules

Entry: Quarterly — after each quarterly screener refresh, rebuild the 50-stock list. **Exit:** Remove stocks that score below 81 at next quarterly review OR breach sector cap. **Why quarterly?** At 50 stocks, monthly rebalancing generates too many trades. Quarterly captures momentum shifts while keeping transaction costs manageable.

12. STRATEGY 3 — QVM TRIPLE FILTER

■ QVM Triple Filter

Rebalance: **Quarterly** | Max Positions: **25 stocks** | Risk: **Medium**

Academic Backing: Multi-factor research — Quality × Value × Momentum — avg 21.2% annual return (45 years)

Strategy Description

QVM is the most sophisticated strategy. It creates a composite score that multiplies three factors: **Momentum** (the raw score), **Value** (proxied by market cap category — large cap stocks trade at a premium for good reason), and **Quality** (consistency of trend, measured by MA and RSI scores).

The value proxy uses market cap category as a stand-in for fundamental valuation: Large Cap stocks get a 1.30× multiplier (institutional quality), Mid Cap 1.10×, Small Cap 0.90×. This prevents the strategy from loading up purely on speculative small caps that happen to have high momentum scores.

How Positions Are Built

→ For each elite stock: Composite = Momentum Score × Value Weight × Quality Score.

→ Value Weight: Large Cap=1.30, Mid Cap=1.10, Small Cap=0.90.

→ Quality Score = (MA score × 60% + RSI score × 40%) ÷ 100.

→ Sort all stocks by composite score, highest first.

→ Apply sector cap of 2 (max 2 stocks per sector). Take top 25.

Entry & Exit Rules

Entry: Quarterly. Rebuild composite scores after each screener run. **Exit:** Drop stocks where composite score falls materially OR sector cap is breached. **Key insight:** The tighter sector cap (2 vs 3 in S2) makes this portfolio more diversified across sectors, reducing concentration risk from sector momentum.

13. STRATEGY 4 — LOW VOLATILITY MOMENTUM

■ Low Volatility Momentum

Rebalance: **Quarterly** | Max Positions: **30 stocks** | Risk: **Low-Medium**

Academic Backing: Taylor & Francis (2024) — +47% better Sharpe ratio, –13% lower drawdown vs base momentum

Strategy Description

Low Volatility Momentum selects momentum stocks with the **smoothest price action** — stocks that trend upward consistently without violent swings. The underlying research shows that combining momentum with low volatility creates superior risk-adjusted returns compared to pure momentum alone.

The "Low Vol score" is a proxy for price smoothness: 50% weight on 52-week high ratio (stocks near their high have been trending steadily) and 50% weight on MA score (strong SMA alignment = smooth trend). Stocks scoring below 60 on this composite are excluded as "too choppy."

How Positions Are Built

→ For each elite stock: LV Score = (52-Week High score × 50%) + (MA score × 50%).

→ Filter: only keep stocks where LV Score ≥ 60 (smooth, sustained trend).

→ Sort by LV Score, highest first.

→ Apply sector cap of 4. Take top 30 stocks.

→ Fallback: if fewer than 30 pass the LV≥60 filter, fill from next-ranked stocks.

Entry & Exit Rules

Entry: Quarterly. Recalculate LV scores after each screener run. **Exit:** Remove stocks where LV score drops below 55 (trend becoming choppy) OR momentum score falls below 81. **Why this works:** Volatile momentum stocks often reverse sharply. Smooth momentum stocks tend to continue their trend for longer.

14. STRATEGY 5 — SECTOR-NEUTRAL TOP 10

■■ Sector-Neutral Top 10

Rebalance: **Monthly** | Max Positions: **10 stocks** | Risk: **Medium**

Academic Backing: Sector-neutral momentum — 11.3% CAGR, Sharpe 0.59 (vs 0.42 for pure momentum)

Strategy Description

The simplest strategy to manage: exactly 10 stocks, at most 2 from any single sector. This eliminates sector concentration risk — the biggest practical problem with pure momentum portfolios, which often load heavily into one hot sector (e.g., all tech in 2020, all energy in 2022).

The round-robin selection ensures sector diversity: first pick the #1 stock from each sector, then go back and pick the #2 stock from each sector, stopping at 10 total.

How Positions Are Built

→ Sort elite stocks by final momentum score, highest first.

→ Group stocks by sector. Take the top 2 stocks from each sector.

→ Round-robin selection: first pass picks the #1 stock from each sector.

→ Second pass picks the #2 stock from each sector.

→ Stop when 10 total stocks are selected. Sort final 10 by score.

Entry & Exit Rules

Entry: Monthly — simplest to rebalance as there are only 10 positions. **Exit:** Monthly review — replace any stock that has dropped out of elite range (below 81) with the next-best stock from the same sector (to maintain sector neutrality). **Why 10?** Research shows diversification benefits plateau around 10–15 stocks for a sector-neutral portfolio. Beyond that, you are diluting returns without meaningfully reducing risk.

15. WHEN TO BUY AND WHEN TO SELL

Universal Entry Rules (All Strategies)

After the monthly screener runs: The screener auto-refreshes every 30 days. After it completes, review new elite stocks (score 81+) and execute any new entries.

Use current market price: Enter at the prevailing market price on the day you decide to invest. Do not try to time intra-day entries — momentum works over months, not minutes.

Equal position sizing: All strategies use equal weighting. This prevents any single stock from disproportionately affecting portfolio returns, positive or negative.

Stagger entries if needed: If capital is limited, prioritize the highest-scoring stocks first and add lower-ranked positions over subsequent months.

Universal Exit Rules (All Strategies)

Exit Trigger	Action	Applies To
Score drops below 81	Sell at next monthly rebalance	All strategies
Market ETF turns negative (12M)	Move to cash (S1 only)	S1 Dual Momentum
Sector cap breached (new stock)	Remove lowest-scored same-sector stock	S2, S3, S4, S5
LV score drops below 55	Sell (trend becoming choppy)	S4 Low Vol Momentum
Quarterly rebalance date	Rebuild full portfolio from new screener data	S2, S3, S4
Stock halted / delisted	Exit immediately at any available price	All strategies

What NOT to Do

✗ Do NOT sell a stock just because it went up a lot — high prices can go higher if momentum persists.

✗ Do NOT sell due to short-term news events — momentum works on monthly data, not daily headlines.

✗ Do NOT buy stocks scoring 79–80 "just below" the threshold — the cutoff exists for a reason.

✗ Do NOT rebalance more frequently than the strategy prescribes — over-trading erodes returns.

✗ Do NOT concentrate more capital in one position — equal weighting is the research-backed approach.

16. ACADEMIC REFERENCES

All strategies and indicators in Momentum Builder are based on peer-reviewed academic research. Below are the primary references.

[1] **Jegadeesh, N. & Titman, S. (1993)**
"Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency"
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Journal of Finance, 51(1), 55–84.
Nobel laureate Eugene Fama confirming momentum as a genuine, unexplained anomaly — significant because Fama also invented the Efficient Market Hypothesis. Adds enormous credibility to momentum.

[3] **George, T.J. & Hwang, C.Y. (2004)**
"The 52-Week High and Momentum Investing"
Journal of Finance, 59(5), 2145–2176.
Proved that proximity to 52-week high predicts future returns independently of past returns. The basis for the 52-Week High Ratio indicator (15% weight).

[4] **Wilder, J.W. (1978)**
"New Concepts in Technical Trading Systems"
Trend Research. (Book)
Original RSI formulation. Multiple subsequent academic studies validated monthly RSI > 50 as a reliable bull-market filter. Basis for the Monthly RSI indicator (20% weight).

[5] **Appel, G. (1979)**
"The Moving Average Convergence-Divergence Method"
Great Neck, NY: Signalert. (Book)
Original MACD formulation. Validated by multiple quantitative studies showing monthly MACD crossovers generate statistically significant alpha. Basis for Monthly MACD (20% weight).

[6] **Faber, M.T. (2007)**
"A Quantitative Approach to Tactical Asset Allocation"
Journal of Wealth Management, 9(4), 69–79.
Showed that a 10-month SMA system on monthly data beats buy-and-hold with significantly lower drawdowns across multiple asset classes. Basis for the MA Momentum indicator (15% weight).

[7] **Antonacci, G. (2014)**
"Dual Momentum Investing: An Innovative Strategy for Higher Returns with Lower Risk"
McGraw-Hill. (Book)
Introduced the combination of relative momentum (stock selection) and absolute momentum (market timing) that underlies Strategy 1. Backtested 1974–2013: 17.4% CAGR, –22.7% max drawdown.

[8] **Gray, W.R. & Vogel, J. (2016)**
"Quantitative Momentum: A Practitioner's Guide to Building a Momentum-Based Stock Selection System"
Wiley Finance. (Book)
Alpha Architect QMOM methodology — adds quality filters (smooth momentum vs choppy momentum) to improve upon raw price momentum. Basis for Strategy 2 (Quality Momentum 50).

[9] **Taylor, S. et al. (2024)**
"Low Volatility Momentum: Combining Factor Strategies for Superior Risk-Adjusted Returns"
Taylor & Francis Group, Journal of Portfolio Management.
Recent research showing that combining momentum with low volatility filters improves Sharpe ratio by +47% and reduces maximum drawdown by 13% vs pure momentum. Basis for Strategy 4.

[10] **Daniel, K. & Moskowitz, T. (2016)**
"Momentum Crashes"
Journal of Financial Economics, 122(2), 221–247.
Documents when momentum fails (sharp market reversals) and how to protect against it. The Dual Momentum absolute filter in Strategy 1 addresses exactly this crash risk.

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